The Most Informative Boolean Function Conjecture Holds for $n = 4$

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Abstract

Let X be uniform on $\{0,1\}^n$ and let Y be the output of a memoryless binary symmetric channel with crossover probability α on input X. Courtade and Kumar conjectured that every Boolean function $f : \{0,1\}^n \to \{0,1\}$ satisfies $I(f(X);Y) \le 1 - h(\alpha)$, where h is the binary entropy function: no one-bit quantization of X is more informative about the noisy observation than a single coordinate. We prove the conjecture for $n = 4$ and every $\alpha \in (0, \frac{1}{2})$, with dictators and anti-dictators the unique maximizers on $[0.001, 0.495]$. To our knowledge this is the first rigorous full-noise-range verification for arbitrary Boolean functions at a fixed n. The proof is computer-assisted but self-contained: two short analytic lemmas handle neighborhoods of the degenerate endpoints $\alpha \to 0$ and $\alpha \to \frac{1}{2}$, with their finitely many per-class hypotheses verified in exact rational arithmetic, and validated interval arithmetic certifies the strict inequality on the compact middle range, class by class over the 222 NPN equivalence classes of 4-variable Boolean functions. An explicit covering lemma assembles the regimes. All computations are reproducible from the public repository.

I. Introduction

How much can a single bit reveal about a noisy vector? Let $X \sim \text{Uniform}(\{0,1\}^n)$ and let Y be obtained from X by passing each coordinate independently through $\mathsf{BSC}(\alpha)$, $\alpha \in (0, \frac{1}{2})$. Every Boolean function $f : \{0,1\}^n \to \{0,1\}$ is a one-bit quantization of the clean data, and data processing alone bounds its relevance to the noisy observer only by $I(f(X);Y) \le \min\{1,\, n(1 - h(\alpha))\}$. Courtade and Kumar [1] conjectured the far sharper

$$I(f(X);Y) \;\le\; 1 - h(\alpha) \tag{1}$$

Code and verification artifacts: https://github.com/gokhanmergen/information-theory-problems.

AI use: the proofs, code, and initial draft were generated by an AI system (Claude Fable 5, Anthropic), adversarially audited by GPT-5 Codex (OpenAI), and validated by Gemini 3.5 Flash (Google) and ChatGPT 5.6 Sol (OpenAI); the human authors are responsible for the content. See the Contribution and AI-Use Statement.

for every Boolean $f : \{0,1\}^n \to \{0,1\}$, with the dictator functions $f(x) = x_i$ and their complements (anti-dictators) achieving equality: the most informative one-bit summary is a single coordinate. The question arose from CEO-style distributed inference problems [1], and (1) is a fundamental limit on the usefulness of a hard decision, made from clean data, to a remote observer who sees only noise-corrupted data. Beyond its operational content, the conjecture has become a benchmark for the interaction of information theory with the analysis of Boolean functions: a one-line, finite-dimensional statement that has resisted hypercontractivity, Fourier perturbation, and isoperimetric methods for over a decade.

A. Related work

The conjecture remains open in general. Among peer-reviewed results: it holds in the high-noise regime $\alpha \in [\frac{1}{2} - \delta, \frac{1}{2})$ for an inexplicit absolute constant δ [2]; improved global bounds short of (1) appear in [3]; the conjecture is equivalent to a symmetrized Li–Médard conjecture [6]; and a differential-equation reformulation reduces the balanced case to a finite-dimensional functional inequality [7]. Yu [4] established local optimality of dictators and, combining it with earlier bounds and computer-assisted estimates, confirmed the balanced conjecture for the explicit correlation range $\rho = 1 - 2\alpha \in [0, 0.914]$, uniformly in n. Among preprints, Javanmard and Woodruff [5] sharpened the high-noise entropy expansion to its optimal order and proved a coordinate-wise form of (1) for all Boolean functions, again with inexplicit constants delimiting the proven range. Two claimed full proofs were later withdrawn by their authors [8], [9]. As numerical evidence, Courtade and Kumar [1, Sec. II-B] enumerated compressed Boolean functions through $n = 7$ and evaluated their mutual information; separately, their implementation of Algorithm II.1 sampled a related inequality on a noise grid of spacing 0.001 [1, Sec. II-D]. Neither computation proves (1) on any noise interval.

For arbitrary (including unbalanced) Boolean functions at a fixed n, no proof covering the entire range $(0, \frac{1}{2})$ appears to have been recorded: the high-noise theorems have inexplicit constants, so they cannot be concatenated with grid computations; and the low-noise limit — where (1) degenerates to equality for every balanced f at $\alpha = 0$ — has no general theorem at all.

B. Contribution

Theorem 1: For $n = 4$, inequality (1) holds for every $\alpha \in (0, \frac{1}{2})$ and every Boolean function f. Moreover, for $\alpha \in [0.001, 0.495]$, equality holds if and only if f is a dictator or an anti-dictator.

The endpoint regimes are exactly where naive certification must fail: the gap $g_f(\alpha) := 1 - h(\alpha) - I(f(X); Y)$ vanishes as $\alpha \to \frac{1}{2}$ for every f, and as $\alpha \to 0$ for every balanced f. Our two endpoint lemmas isolate the finite data controlling (1) there.

a) Endpoint regimes: Edge-boundary profiles and hypercube edge-isoperimetry control the conjecture near $\alpha = 0$; level-one Fourier weight and spectral ℓ_1-mass control it near $\alpha = \frac{1}{2}$. Each endpoint reduces to finitely many rational inequalities, verified exactly.

b) Middle range: Validated interval arithmetic certifies strict inequality on the compact middle range, class by class (Section V).

c) Complete coverage: Lemma 4 assembles the regimes into a gap-free cover of $(0, \frac{1}{2})$. The scheme is dimension-generic; Section VII reports its state at $n = 5$.

Remark 1 (Scope of the uniqueness claim): The interval computation certifies strict inequality for every non-dictator class on $[0.001, 0.495]$, which yields the equality characterization there. The endpoint lemmas establish $g_f \geq 0$ without a uniform separating margin, so uniqueness on $(0, 0.001) \cup (0.495, \frac{1}{2})$ is expected but not certified here; no non-dictator equality case can occur on the certified range, and we know of no mechanism producing one in the endpoint slivers.

II. Setup, symmetry reduction, and class accounting

Throughout, $h(u) = -u \log_2 u - (1-u) \log_2(1-u)$ with the convention $0 \log_2 0 := 0$; $\rho = 1 - 2\alpha$. Write $q_1 = \Pr[f(X) = 1] = |f^{-1}(1)|/16$ and $q_0 = 1 - q_1$; neither depends on α. Since Y is uniform on $\{0,1\}^4$,

$$g_f(\alpha) = H\big(f(X), Y\big)(\alpha) - h(\alpha) - 3 - H(q_1),$$

and the 2×16 joint probabilities are the exact polynomials

$$p_{v,y}(\alpha) = 2^{-4} \sum_{d=0}^{4} c_{v,y,d}\, \alpha^d (1-\alpha)^{4-d}, \qquad c_{v,y,d} = \#\{x : \ f(x) = v,\ d_H(x,y) = d\},$$

where d_H is Hamming distance; this uses that, given $Y = y$, the posterior of X is the product BSC kernel centered at y (Bayes' rule with uniform input and a symmetric channel).

TABLE I

Disposition of the 222 NPN classes.

Classes	Method	Conclusion
1 constant	analytic: $I = 0$	$g_f = 1 - h(\alpha) > 0$
1 dictator	analytic: $I = 1 - h(\alpha)$	$g_f \equiv 0$ (equality)
220 remaining	Lemmas 2, 3; Proposition 1	$g_f > 0$ (middle), $g_f \geq 0$ (endpoints)

$I(f(X);Y)$ is invariant under permutations of the coordinates, flips of input coordinates, and complementation of the output, so it suffices to prove (1) for one representative of each NPN equivalence class. For $n = 4$ there are 222 classes [10], a count our enumeration reproduces, with orbit sizes summing to 2^{16}. The accounting used throughout is given in Table I. Every computer-assisted step below covers the same 220 classes; the interval certification additionally re-verifies the constant class (whose zero $c_{v,y,d}$ counts drop out of the entropy sum by the $0 \log_2 0$ convention), which is why its logs report 221.

III. The low-noise lemma

For $y \in \{0,1\}^4$ let b_y be the number of Hamming neighbors x of y with $f(x) \neq f(y)$, and set $c_1 = \frac{1}{16} \sum_y b_y$ (so $16c_1 = 2B$ with B the edge boundary of $f^{-1}(1)$) and $c_2 = \frac{1}{16} \sum_y b_y \log_2 b_y$ (terms with $b_y = 0$ vanish by the $0 \log_2 0$ convention; $b_y \leq 4$ gives $c_2 \leq 2c_1$).

Lemma 1 (code-verified enumeration): Over the balanced NPN classes of Q_4, the minimum edge boundary of a non-dictator class is 12; dictators attain the edge-isoperimetric minimum 8, uniquely among balanced sets.

The uniqueness of subcubes at the edge-isoperimetric minimum is classical [11], [12], [13], [14]; the value 12 for the best balanced non-dictator is an exhaustive computation over the balanced classes, recorded in the repository (it is not claimed to follow from the isoperimetric theorems).

Lemma 2: Let $\alpha \in (0, 10^{-3}]$, $t = \log_2(1/\alpha)$, $t_0 = \log_2 1000$, $\gamma = (1-10^{-3})^3$. If f is balanced and $t_0(\gamma c_1 - 1) \geq \log_2 e + c_2$, or f is unbalanced and $10^{-3}\left[t_0 \max(0, 1 - \gamma c_1) + \log_2 e + c_2\right] \leq 1 - H(q_1)$, then $g_f(\alpha) \geq 0$.

Proof: Write $g_f = m_0 - D(\alpha)$ with $m_0 = 1 - H(q_1)$ and $D(\alpha) = h(\alpha) - H(f(X) \mid Y)$. Three elementary steps:

(i) $H(f(X) \mid Y{=}y) = h(\Pr[f(X) \neq f(y) \mid y])$, and $\Pr[f(X) \neq f(y) \mid y] \geq b_y\, \alpha(1-\alpha)^3 =: b_y\beta$ by keeping only the distance-one terms of the posterior kernel. Both quantities are at most $\Pr[X \neq y \mid Y{=}y] = 1-(1-\alpha)^4 \leq 4\alpha \leq \frac{1}{2}$, and h is increasing on $[0, \frac{1}{2}]$, so $H(f(X) \mid Y) \geq \frac{1}{16}\sum_y h(b_y\beta)$.

(ii) $h(u) \geq u\log_2(1/u)$ for $u \in [0,1]$, hence $h(b_y\beta) \geq b_y\beta\,[t - \log_2 b_y]$ using $\beta \leq \alpha$.

(iii) $h(\alpha) \leq \alpha(t + \log_2 e)$, from $(1-\alpha)\ln\frac{1}{1-\alpha} \leq \alpha$.

Combining, and applying $\beta \geq \gamma\alpha$ to the aggregate $c_1 t - c_2$ — which is nonnegative since $c_2 \leq 2c_1$ and $t \geq t_0 > 2$ — while bounding the residual c_2 term with $\beta \leq \alpha$:

$$D(\alpha) \;\leq\; \alpha\big[\,t(1-\gamma c_1) + \log_2 e + c_2\,\big].$$

If f is balanced then $m_0 = 0$ and $c_1 \geq \frac{3}{2}$ by Lemma 1; the t-coefficient is negative and $t \geq t_0$ gives the stated criterion. If f is unbalanced, αt is increasing on $(0, 1/e)$, so D is maximized at $\alpha = 10^{-3}$, giving the stated criterion against $m_0 > 0$. ■

All 220 classes satisfy the applicable criterion, verified in exact rational arithmetic with two-sided rational enclosures of $\log_2 3, \log_2 5, \log_2 7, \log_2 11, \log_2 13, \log_2 e, \log_2 1000$ (each proved by integer exponentiation or a rational series, and used in the unfavorable direction). The minimum slack over all classes is $1680/546377 \approx 3.1 \times 10^{-3}$, attained by class 0x1669 ($q_1 = 7/16$); no class is borderline.

IV. The high-noise lemma

Fourier conventions: for $S \subseteq \{1, \ldots, 4\}$ let

$$\chi_S(y) = (-1)^{\sum_{i\in S} y_i}, \qquad \widehat{\mathbf{1}_f}(S) = 2^{-4}\sum_x \mathbf{1}_f(x)\chi_S(x),$$

so $\mathbf{1}_f = \sum_S \widehat{\mathbf{1}_f}(S)\chi_S$ and the noise operator satisfies $T_\rho\mathbf{1}_f = \sum_S \widehat{\mathbf{1}_f}(S)\rho^{|S|}\chi_S$ with $T_\rho\mathbf{1}_f(y) = \Pr[f(X) = 1 \mid Y = y]$. Parseval gives $\sum_S \widehat{\mathbf{1}_f}(S)^2 = 2^{-4}\sum_x \mathbf{1}_f(x)^2 = q_1$ and hence $\sum_{S\neq\emptyset} \widehat{\mathbf{1}_f}(S)^2 = q_1 - q_1^2 = q_0 q_1$. Define $\widetilde{W}_1 = \sum_{|S|=1} \widehat{\mathbf{1}_f}(S)^2/(q_0q_1) \in [0,1]$, $A = \sum_{S\neq\emptyset} |\widehat{\mathbf{1}_f}(S)|$, and $q_{\min} = \min(q_0, q_1)$. Constants have $q_{\min} = 0$ and are excluded; they were handled in Section II.

Lemma 3: Let $\rho_0 = 10^{-2}$. If $\rho_0 A/q_{\min} \leq \frac{1}{2}$ and

$$\big[\widetilde{W}_1 + \rho_0^2(1-\widetilde{W}_1)\big]\big(1 + \tfrac{4}{3}\rho_0 A/q_{\min}\big) \;\leq\; 1,$$

then $g_f(\alpha) \geq 0$ for all $\rho \in (0, \rho_0]$.

Proof: Write $p(v,y) = q_v 2^{-4}(1+\varepsilon_{v,y})$. Since $p(1,y) = 2^{-4}T_\rho \mathbf{1}_f(y)$ and $p(0,y) = 2^{-4}(1 - T_\rho \mathbf{1}_f(y))$,

$$\varepsilon_{1,y} = \frac{T_\rho \mathbf{1}_f(y) - q_1}{q_1}, \qquad \varepsilon_{0,y} = -\frac{T_\rho \mathbf{1}_f(y) - q_1}{q_0},$$

and $|T_\rho \mathbf{1}_f(y) - q_1| \le \sum_{S\neq\emptyset} |\widehat{\mathbf{1}_f}(S)|\rho^{|S|} \le \rho A$ for $\rho \le 1$, so $|\varepsilon_{v,y}| \le \rho A/q_{\min} =: \varepsilon_{\max}$. By Parseval,

$$\chi^2 := \sum_{v,y} q_v 2^{-4}\varepsilon_{v,y}^2 = \frac{\sum_{S\neq\emptyset} \widehat{\mathbf{1}_f}(S)^2\rho^{2|S|}}{q_0 q_1} \le \rho^2\big[\widetilde{W}_1 + \rho^2(1-\widetilde{W}_1)\big],$$

using $\rho^{2|S|} \le \rho^4$ for $|S| \ge 2$. For $|\varepsilon| \le \frac{1}{2}$, $(1+\varepsilon)\ln(1+\varepsilon) \le \varepsilon + \frac{\varepsilon^2}{2} + \frac{2}{3}|\varepsilon|^3$: the series $(1+\varepsilon)\ln(1+\varepsilon) = \varepsilon + \sum_{k\ge 2}\frac{(-1)^k\varepsilon^k}{k(k-1)}$ is, for $\varepsilon \ge 0$, alternating with decreasing terms after $\varepsilon^2/2$ (negative leading tail term), and for $\varepsilon \in [-\frac{1}{2}, 0)$ all terms are positive with tail at most $\frac{|\varepsilon|^3}{6}\cdot\frac{1}{1-|\varepsilon|} \le \frac{|\varepsilon|^3}{3}$. Since $\sum_{v,y} q_v 2^{-4}\varepsilon_{v,y} = 0$ (both marginals are exact), summing gives $I(f(X);Y)\ln 2 \le \frac{\chi^2}{2}(1+\frac{4}{3}\varepsilon_{\max})$. Finally $1 - h(\alpha) = \frac{1}{\ln 2}\sum_{k\ge 1}\frac{\rho^{2k}}{2k(2k-1)} \ge \frac{\rho^2}{2\ln 2}$ with all series terms positive. Because $1 - \widetilde{W}_1 \ge 0$, both bracketed factors in the hypothesis are nondecreasing in ρ, so verification at ρ_0 suffices for all $\rho \in (0, \rho_0]$. ■

All 220 classes pass, including all balanced ones, with exact dyadic Fourier data; no external high-noise theorem is invoked. The maximum $\widetilde{W}_1$ over non-dictator classes is 52/63 (a dictator altered on a single input), and the minimum slack of the lemma's hypothesis is 0.1416, attained by that same class; no class is borderline.

V. Validated computation on the middle range

Proposition 1: For every one of the 221 NPN representatives (the 220 classes plus the constant class) and every subinterval recorded in the certification runs below, the outward-rounded interval evaluation of g_f encloses the true range of g_f on that subinterval, its infimum is positive, and the recorded subintervals cover $[10^{-3}, 0.0055] \cup [0.005, 0.495]$ together with the exact-rational bridges $[9/10^4, 11/10^4]$ and $[4949/10^4, 4951/10^4]$.

a) Enclosure validity: The proposition rests on five ingredients:

- NPN reduction correctness: Section II; class count and orbit sum checked against [10].
- Polynomial exactness: the integer-coefficient polynomials $p_{v,y}$ are constructed exactly.
- Enclosure properties: each interval operation used — addition, multiplication, division, and logarithm in mpmath.iv at 90-bit precision — is outward-rounded; only these operations are used, and although mpmath labels interval support experimental, this elementary-operation path is deterministic and regression-tested in the repository.

- Adaptive bisection: a subinterval is discharged only when the enclosure's infimum is positive.
- Recorded ledgers for every discharged subinterval.

b) Implementation and runtime: 2029 s and 2.6 s single-threaded (Python 3.14, mpmath 1.3.0, macOS/arm64).

c) Reproducibility: The trusted computing base beyond mpmath is a ∼30-line enclosure routine printed in the repository.

d) Archival status: An immutable archived release with a machine-readable per-subinterval certificate and an independent certificate checker is planned before submission. The audit history, including a repaired floating-point endpoint defect that motivated the two bridges, is in Appendix A.

VI. Covering the noise interval

Lemma 4: Every $\alpha \in (0, \frac{1}{2})$ lies in at least one verified regime, and consecutive regimes overlap on sets of positive measure:

$$(0, \tfrac{1}{2}) \;=\; \underbrace{(0, 10^{-3}]}_{\text{Lem. 2}} \cup \underbrace{[\tfrac{9}{10^4}, \tfrac{11}{10^4}]}_{\text{bridge}} \cup \underbrace{[10^{-3}, 0.0055] \cup [0.005, 0.495]}_{\text{Prop. 1 runs}} \cup \underbrace{[\tfrac{4949}{10^4}, \tfrac{4951}{10^4}]}_{\text{bridge}} \cup \underbrace{[0.495, \tfrac{1}{2})}_{\text{Lem. 3}}.$$

Proof: The union contains $(0, \frac{1}{2})$ by inspection of the rational endpoints; the overlaps $(0, 10^{-3}] \cap [\frac{9}{10^4}, \frac{11}{10^4}]$, $[\frac{9}{10^4}, \frac{11}{10^4}] \cap [10^{-3}, 0.0055]$, $[0.005, 0.0055]$, $[0.4949, 0.495]$, and $[0.495, 0.4951]$ all have positive length. The bridges are certified with exactly represented rational endpoints, eliminating any dependence on the floating-point endpoint handling of the two main runs (Appendix A). ■

Theorem 1 follows: constants and dictators are analytic (Section II); every other class satisfies $g_f \geq 0$ on the endpoint regimes (Lemmas 2, 3) and $g_f > 0$ on the middle range (Proposition 1); and the regimes cover $(0, \frac{1}{2})$ (Lemma 4). The uniqueness statement is Remark 1.

VII. Discussion

The four-regime scheme is dimension-generic. At $n = 5$ there are 616,126 NPN classes — roughly three orders of magnitude more than at $n = 4$ — and the two endpoint lemmas have already been verified for all of them in exact arithmetic (with the low-noise cutover moved to 10^{-4}, as the near-balanced margins shrink); the middle-range certification is embarrassingly parallel by class and is in progress. The genuinely new difficulty as n grows is quantitative: the dictator's isolation gap shrinks (single flips carry input mass 2^{-n}), so the certified middle range must push closer to the endpoints, where the lemmas' margins — edge-isoperimetric at low noise, spectral at high noise — grow more robust. Formal verification (e.g., in Lean) of the two finite endpoint checks is a natural hardening step: the per-class conditions are pure rational arithmetic over explicitly listed data.

Contribution and AI-Use Statement

In accordance with IEEE policy on AI-generated content:

a) Initial generation: The proof strategy, the two endpoint lemmas, all code, and the initial manuscript draft were produced by an AI system (Claude Fable 5, Anthropic) operating on a public open-problems repository.

b) Independent audit and repair: A separate AI system (GPT-5 Codex, OpenAI) performed adversarial audits that identified the endpoint defect and a citation-provenance correction recorded in Appendix A, and repaired the certification script.

c) Validation of the proofs: The endpoint lemma conditions and the manuscript's mathematical claims were independently validated by Gemini 3.5 Flash (Google) and ChatGPT 5.6 Sol (OpenAI).

d) Human responsibility: The human maintainer supervised the process, reviewed the mathematics, and is responsible for the content. AI-generated material appears throughout the manuscript. AI audit does not substitute for independent human verification; readers are invited to check the two lemmas by hand (they are elementary) and to re-run the public code.

Appendix A
Audit History and Validation

The artifact was adversarially audited before being recorded; two findings are material. First, the script used for the two recorded interval runs converted its decimal endpoint strings to binary floating point before intervalization, rounding 0.001 upward by 2.08×10^{-20} and 0.495 downward by 4.44×10^{-18}; the recorded runs therefore covered two microscopically shortened intervals. The repaired script keeps all endpoints as exact rationals until evaluation, and the two rational-endpoint bridges of Lemma 4 independently certify neighborhoods of both seams, so the covering argument does not depend on the original endpoint handling. Second, an earlier internal review proposed an alternative high-noise proof by discarding a fourth-order Taylor remainder; that argument has the wrong inequality direction ($h^{(4)} < 0$ makes the corresponding contribution to mutual information positive) and is not used anywhere: the high-noise regime rests solely on Lemma 3. The hard-coded logarithm enclosures used by the exact-rational checks are themselves proved by integer exponentiation (e.g., $2^a \le n^b$) or a truncated rational series for $\ln 2$, in a self-test that runs before every verification.

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